

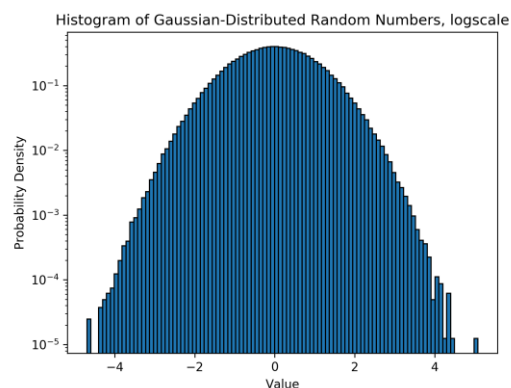
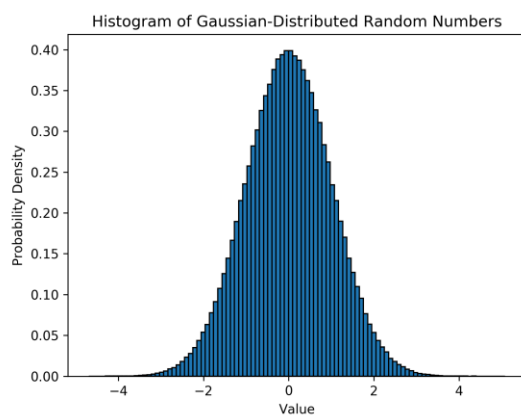
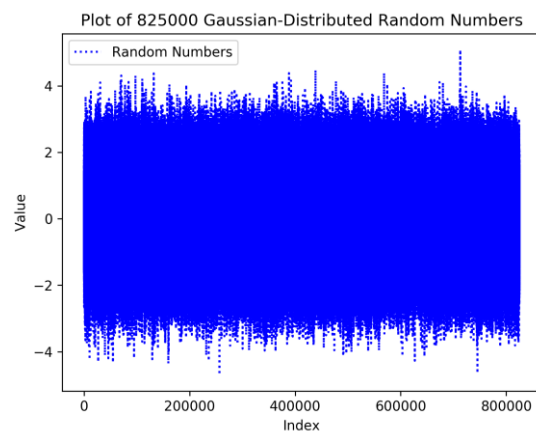
S Hutzler, 16-08-24

Some notes on Simon's foam coarsening data wetfoam2_bub_RH2_0.14_0.503755.dat

I analysed Simon's Surface Evolver produced data set of initially 400 bubbles (liquid fraction 0.14, finite contact angle ...) which undergo coarsening. The foam is followed over a total of 2087 time-steps (iterations) over which 300 bubbles vanish. **Total number of position data points: $412600 \times 2 = 825200$.**

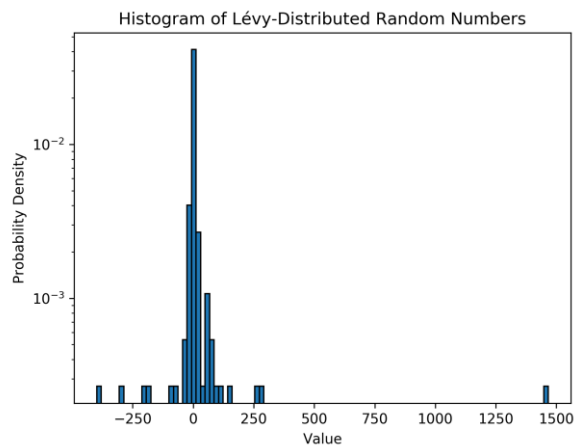
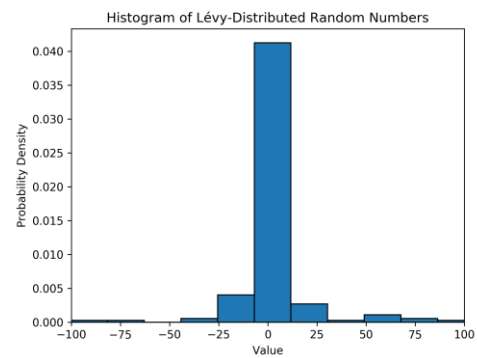
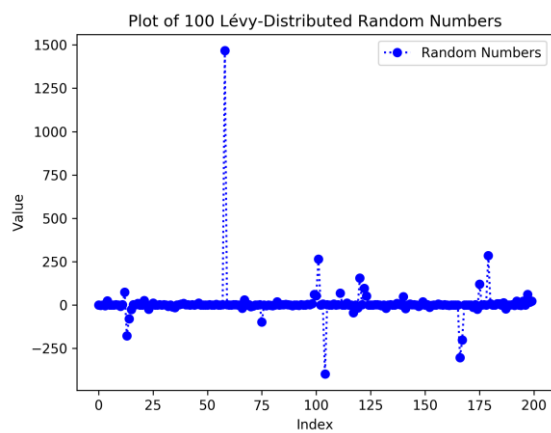
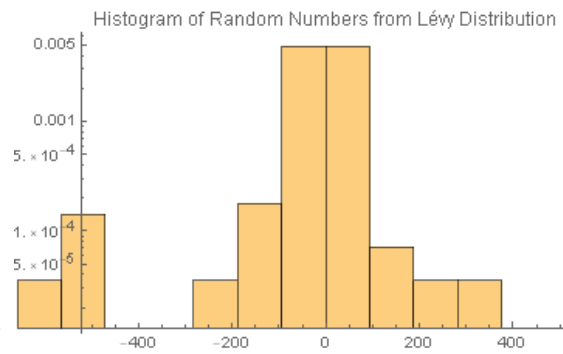
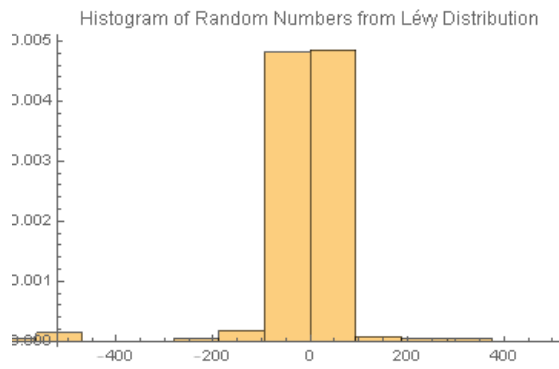
My aim was to study the displacement of individual bubbles in time. A smooth variation should be punctuated by rapid displacements due to T1s. One would thus not expect Gaussian statistics.

Just for reasons of illustration: Computer generated random numbers from Gaussian and non-Gaussian distributions



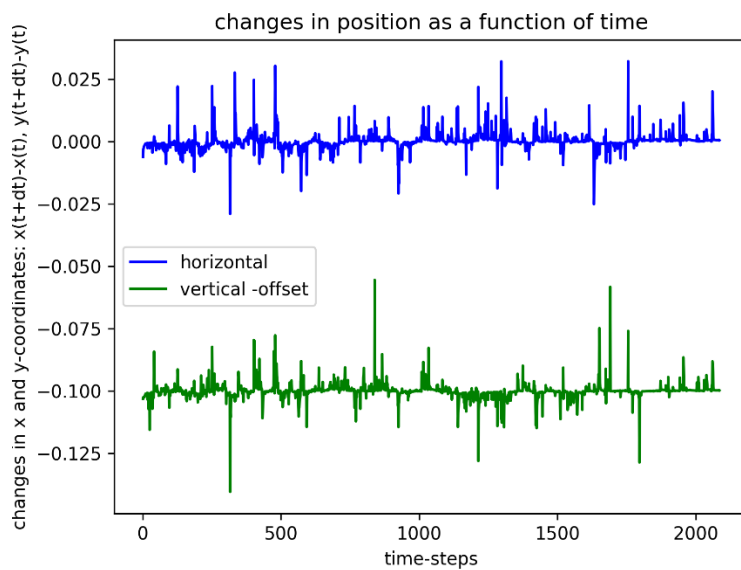
The vertical log-scale reveals the inverted parabola of the Gaussian distribution.

Levy data shows the wide-spread, i.e. non-finite variance (much harder to get nice graphs as the fluctuations are so large)

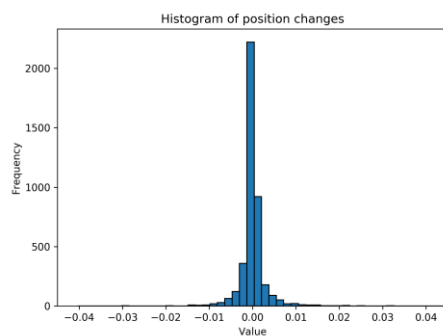


(200 values)

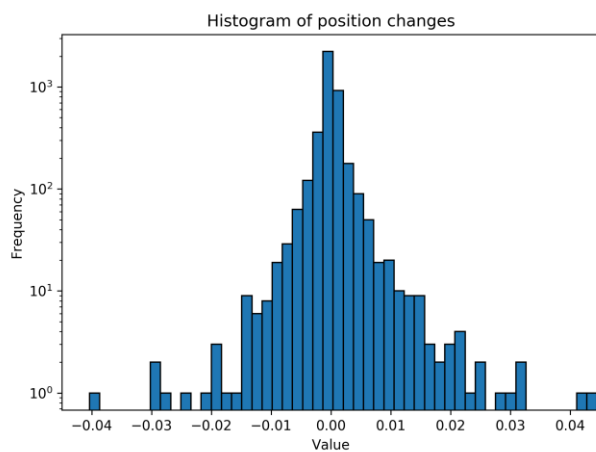
From Simon's data for x-coordinates of one bubble (bubble 1), plot $x(t+1)-x(t)$, and also $y(t+1)-y(t)$, i.e. I consider consecutive changes in the position, $\Delta t=1$



I have computed histograms which describe these fluctuations, I averaged data for changes in x and in y as they show similar statistics.



Same data with vertical axis on log-scale



In econophysics, for stock prices $s(t)$ one defines the return as $R(t,dt) = s(t+dt)/s(t) - 1$. To account for the possibility of constant growth (inflation) one prefers to use the log-return as $r(t,dt) = \ln(s(t+dt)/s(t))$. [In the case of gas diffusion we too would expect displacements to change as small bubbles disappear and large ones get larger.]

A *volatility* is then defined as $|r(t,dt)|$.

Features of stock market data:

Data shows clustering in time of volatility.

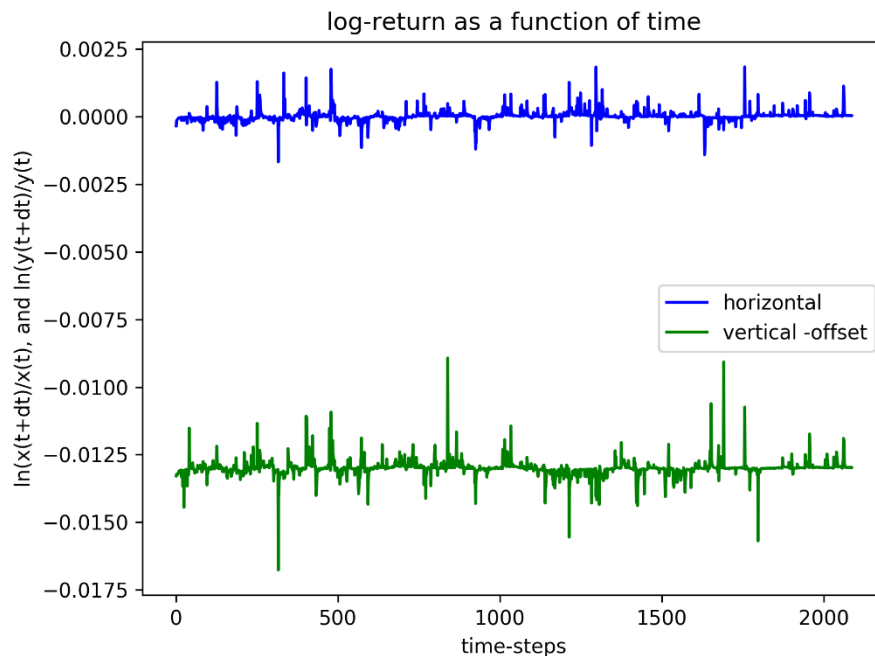
The probability distribution of log-return or probability has fat tails and can be approximated by Levy distributions.

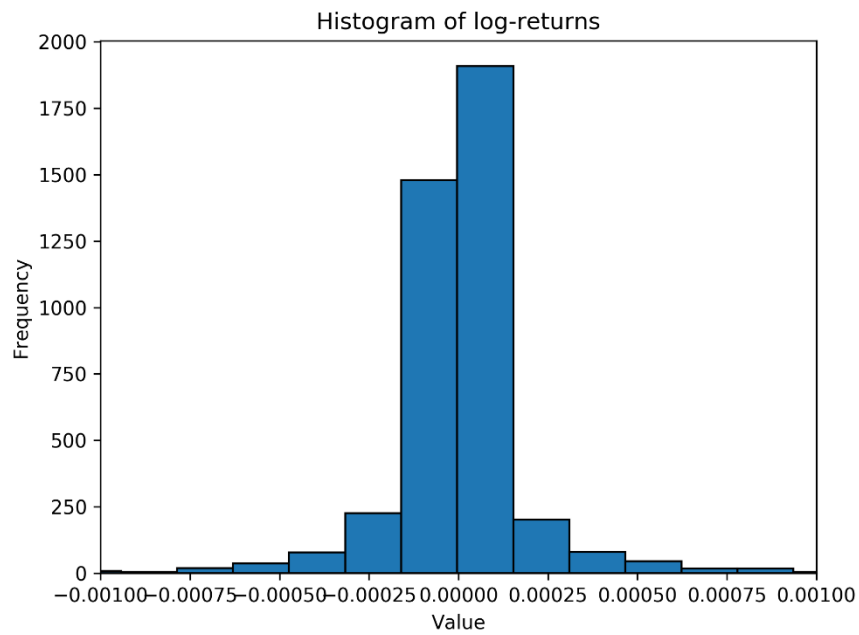
There is a non-linear scaling in dt for the probability distributions for $r(t,dt)$ (diffusion coefficients etc.)

The autocorrelation function $R(dt)$ of the return decays rapidly, those for the log-return shows power law decay.

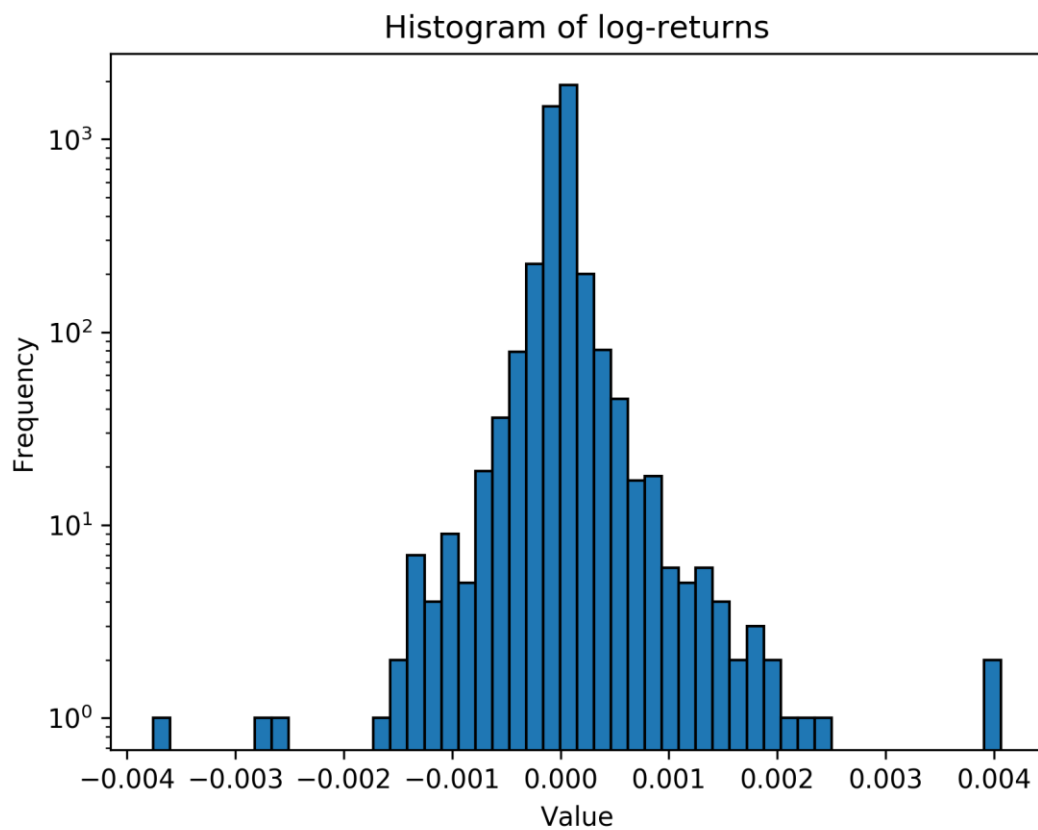
I have started to analyse Simon's data for bubble positions as if it was stock market data.

From Simon's data for x-coordinates of one bubble, plot $x(t+1)/x(t)$ plot $\log(x(t+1)/x(t))$, i.e. I consider consecutive changes in the data, $dt=1$



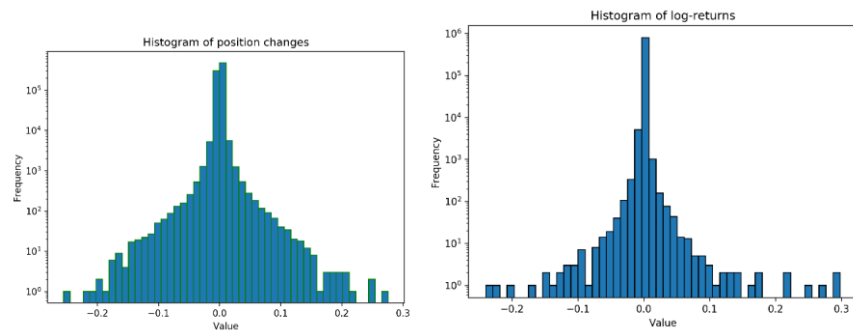
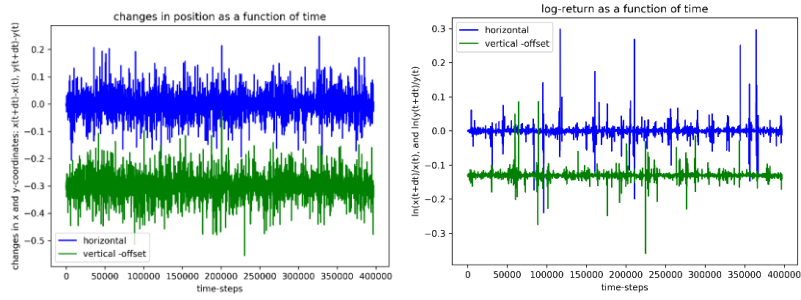


Same data but now on a log scale.

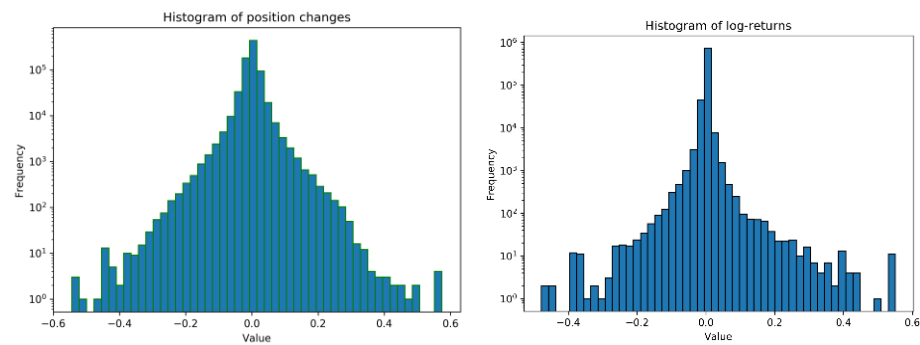
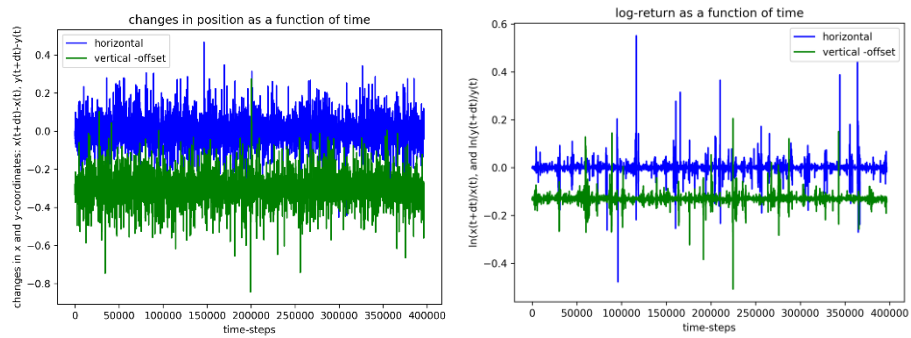


I looked at a number of different bubbles which show similar behaviour. For this reason I have averaged the data obtained from 399 bubbles.

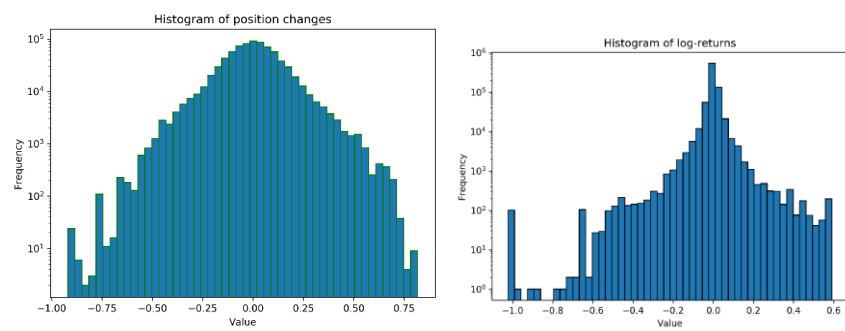
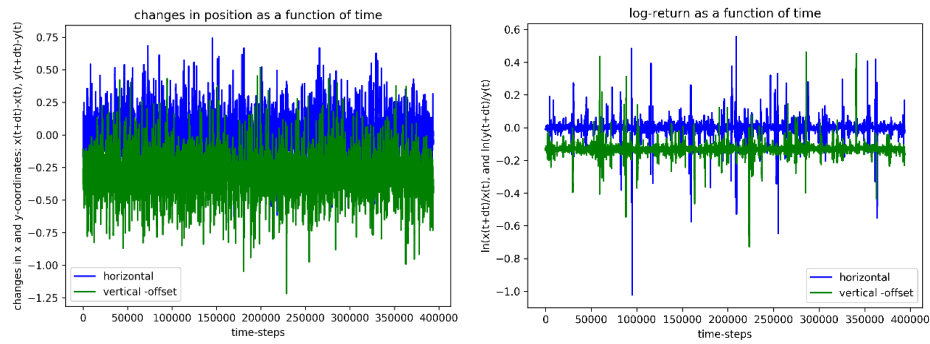
$\Delta t=1$



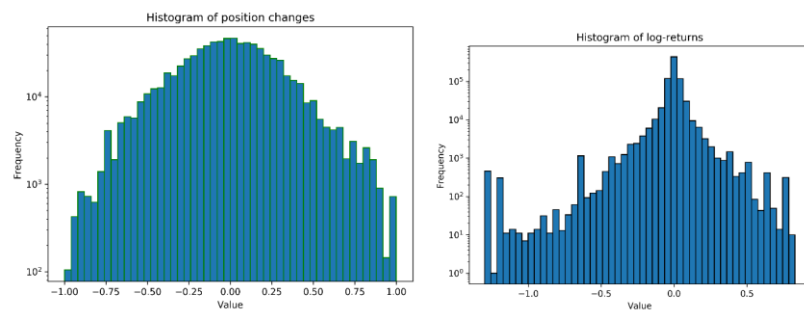
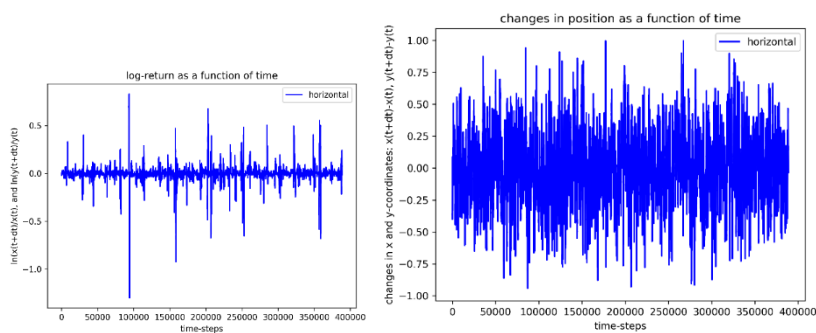
$\Delta t=10$



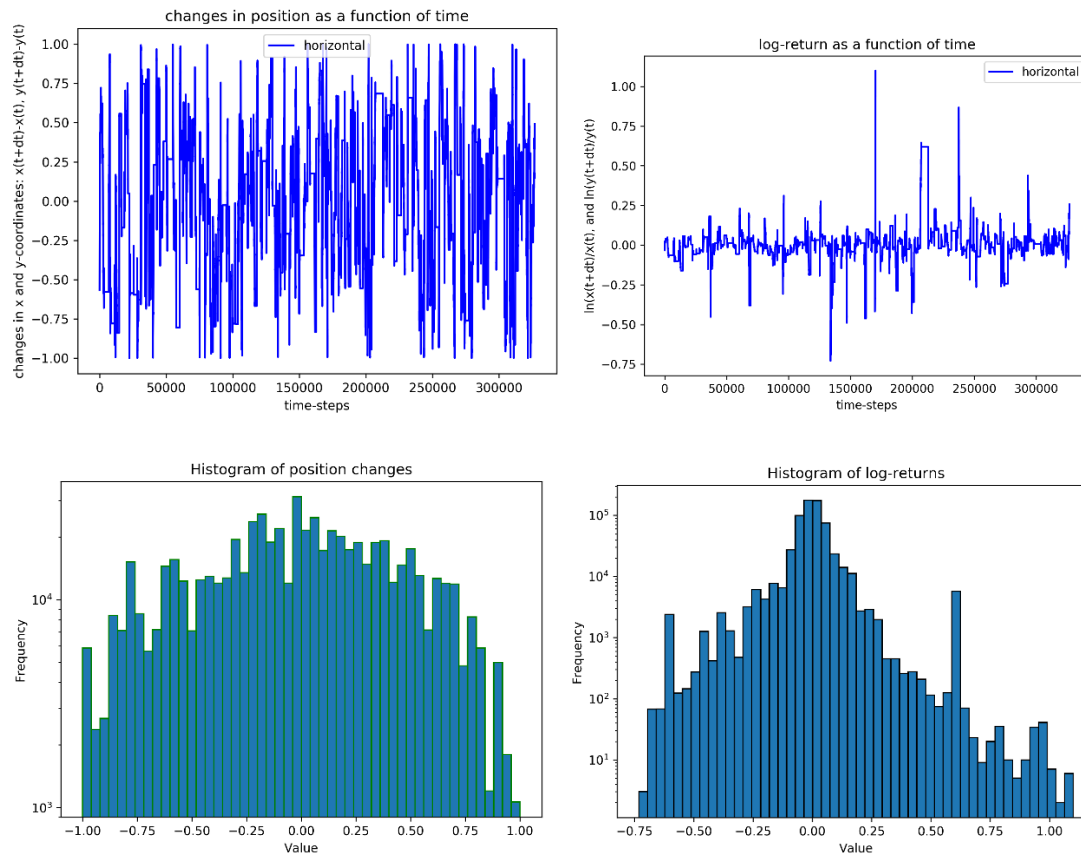
$\Delta t=100$



$\Delta t=300$ (here: for position data show x only)



$\Delta t=1000$ (here: for position data show x only)



One can see that the distributions for increasing Δt from 1 to 1000 approach a Gaussian (visible at least for the position changes); but remember that for $\Delta t = 1000$ we have much less data available.

Suggestions:

It would be interesting to investigate how the above log-return distributions scale with the interval Δt , i.e. the transition to the Gaussian.

Computation of autocorrelation function.

Analysis of data according to number of sides?

Correlation of motion of different bubbles?

Application to experimental ISS data